Madgwick Filter: Teaser

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Recap: Quaternions

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Quaternion Rotation Intuition

$$\boldsymbol{q} = \begin{bmatrix} q_0 \\ \boldsymbol{v} \end{bmatrix} \qquad \text{where} \qquad \boldsymbol{v} = \begin{bmatrix} q_1 \\ q_2 \\ q_2 \end{bmatrix}$$

$$\boldsymbol{q} = \left[\cos\frac{\psi}{2},\ \sin\frac{\psi}{2}\,\boldsymbol{v}^{\mathsf{T}} \right]$$

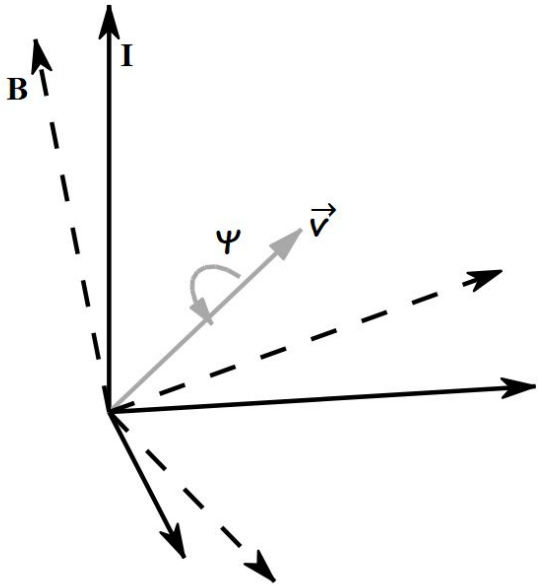

Fig. Axis-angle interpretation of quaternion.

Differential Equations

Newton's second law: $\boldsymbol{F} = m\dfrac{d^2x}{dt^2}$

Euler's equations: $\boldsymbol{I}\dot{\boldsymbol{\omega}} + \omega \times (\boldsymbol{I}\boldsymbol{\omega}) = \boldsymbol{\tau}$

Simple pendulum: $\dfrac{d^2\theta}{dt^2} + \dfrac{b}{m}\dfrac{d\theta}{dt} + \dfrac{g}{L}\sin\theta = 0$

Schrödinger equation: $\dfrac{1}{c^2}\dfrac{\partial^2\phi_n}{\partial t^2} - \nabla^2\phi_n + (\dfrac{mc}{\boldsymbol{\hbar}})^2\phi_n = 0$

Attitude from Angular Rate

$$\dot{\boldsymbol{q}} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} q_0 & q_1 & -q_2 & -q_3 \\ q_1 & q_0 & -q_3 & q_2 \\ q_2 & q_3 & q_0 & -q_1 \\ q_3 & -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} 0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$\dot{\boldsymbol{q}} = \frac{1}{2} \hat{\boldsymbol{q}}_{t-1} \odot \boldsymbol{w}_t$$

$$\boldsymbol{q}_t = \hat{\boldsymbol{q}}_{t-1} + \dot{\boldsymbol{q}} dt$$

Attitude from Vector Observations

$$\boldsymbol{v}^B = \boldsymbol{q}^* \odot \boldsymbol{v}^I \odot \boldsymbol{q}, \text{ ideal vector in body frame,}$$

$$\tilde{\boldsymbol{v}}^B \approx \boldsymbol{q}^* \odot \boldsymbol{v}^I \odot \boldsymbol{q}, \text{ vector measurement, and}$$

$$\tilde{\boldsymbol{v}}^B = \boldsymbol{v}^B + \text{noise.}$$

$$\boxed{\min_{\boldsymbol{q} \in \mathbb{H}} \boldsymbol{f}(\boldsymbol{q}, \tilde{\boldsymbol{v}}^B, \boldsymbol{v}^I) = \boldsymbol{v}^B - \tilde{\boldsymbol{v}}^B}$$

White Board

Thank you !!